



ULAB

UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Course Title : ICT For Development

Course Code : MSJ 3261

Section : 1

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ID : 222012034

Date : 02/12/2024

Abstract

Information and Communication Technology (ICT) has become the foundation of sustainable development through economic growth, social inclusion and environmental protection. ICT has become one of the tools of development in today's world. In Bangladesh, ICT plays an important role in achieving the Sustainable Development Goals (SDGs), especially in the implementation of initiatives such as "Digital Bangladesh". ICT plays an important role in socio-economic development in developing countries like Bangladesh. However, despite significant progress, the country faces significant challenges, including infrastructural deficits, digital divide, lack of skilled professionals and policy flaws. This paper critically examines the implications of the digital divide in Bangladesh, highlighting disparities in access to digital tools between urban and rural populations, socio-economic class and gender. It also explores the opportunities ICT offers, including expanding Internet access, government initiatives and the potential of emerging technologies. The paper concludes by presenting actionable recommendations to strengthen ICT infrastructure, enhance digital literacy and promote inclusion. By addressing this challenge, Bangladesh can fully exploit the potential of ICT for sustainable development and elimination of existing inequalities. Basically, how ICT plays a role in sustainable development and creation of skilled manpower is highlighted here. Developing countries like Bangladesh, who are slow to adopt new technologies, seem like challenges, but ICT has opened the door of immense possibilities. This requires digital signatures to be converted into skilled manpower.

ICT for Sustainable Development in Bangladesh: Challenges, Opportunities and Future Prospects

Introduction

Information and communication technology has become a powerful tool for economic prosperity, social development and environmental protection in the modern world. ICT plays a very important role for a developing country like Bangladesh in ensuring good governance, achieving equitable growth and empowering the people. The agenda was unveiled with ICT at the center to create a Digital Bangladesh, with the aim of transforming the country into a knowledge-based economy by 2041. Doing. The role of technology in eradicating poverty, improving living standards and addressing global challenges is unique. Bangladesh has made significant progress in the field of information and communication technology in the last two decades. But a question remains when compared with developed countries. Broadband expansion, modernization of public services, and programs to encourage technology-based entrepreneurship have spurred economic activity and delivery. The rise of freelancing, software exports and mobile banking platforms are examples of how ICT is reshaping industries and creating new opportunities. ICT is also playing an important role in social progress such as e-learning platforms that expand the education system, telemedicine that reaches the underprivileged population with services and e-governance tools that ensure transparency and accountability. Despite these advancements, significant challenges remain. Many rural areas still lack reliable internet access, creating a digital divide that disproportionately affects the poor, women, and marginalized groups. The high cost of devices, inadequate digital literacy, and insufficient investments in cutting-edge technologies further exacerbate these disparities. Cybersecurity threats and outdated policies, such as the ICT Act 2002, also hinder the safe and efficient use of ICT.

The role of ICT in fostering sustainable development in Bangladesh, exploring its contributions, challenges, and potential future pathways is very significant. Particular attention is given to the "digital divide" and its implications for inclusive growth. The analysis also underscores the need for robust policies and reforms to maximize the benefits of ICT while addressing the challenges. As Bangladesh progresses toward its development goals, ICT's integration into all sectors will be pivotal in ensuring long-term, inclusive, and equitable growth.

ICT in Bangladesh and Sustainable Development Goals (SDGs)

The SDGs are a set of 17 global development goals adopted by the United Nations, which aim to ensure sustainable development. And ICT plays an important role in this sustainable development. ICT initiatives in Bangladesh are aligned with several SDGs, including:

SDG 4: Quality Education: ICT has increased access to education through platforms such as Shikhhok Batayan and e-learning systems, especially during the Covid-19 pandemic. Digital classrooms and online resources aim to reduce educational disparities, especially for rural areas. Through which even a student from a remote area can ensure quality education.

SDG 8: Decent work and economic growth: The ICT sector plays an important role in the economy of Bangladesh. The development of ICT has led to the creation of employment in software development and digital freelancing among others. As a result, a large section of the country's youth is turning to freelancing and earning a lot of foreign exchange, which is contributing significantly to the country's economy. The government aims to earn \$5 billion from ICT exports by 2025.

SDG 9: Industry, Innovation, and Infrastructure: Expansion and innovation of ICT infrastructure including broadband connectivity and hi-tech parks have made the industrial sector more dynamic. The development of smart cities reflects the progress of sustainable urbanization. Today ICT has a direct and indirect role in every field of industrialization. ICT plays a role everywhere from the production of a product to its delivery to the customer. Use of ICT everywhere from employee payroll to supervision. The work done by robots in industrial factories has also become possible due to the development of ICT. From infrastructural development planning to entire roadmaps are created using ICiri tools. ICT also plays a role in the desktop research we do in some research areas. Development has become more dynamic as ICT has been linked to industry, innovation and infrastructure development.

SDG 10: Reduced inequality: ICT can be a great tool in reducing inequality. Discrimination does not mean only men and women. Technological inequality can also be of many forms such as a person in a village gets more benefits from the use of ICT than a person in a city. In

addition, in many cases we see that some people have access to various websites but others cannot use them even if they have rights. Such discrimination can be eliminated only through the use of ICT tools. Social media plays an important role in this regard. The role of ICT in reducing inequality in mobile banking, microcredit, online business etc. is very important. Just as a city person gets benefits in these areas, a village person also gets benefits in the same way. ICT also plays an important role in reducing gender inequality. From online business to social media, women are everywhere. Which plays an important role in ensuring the rights of women.

SDG 16: Peace, justice and strong institutions: E-governance initiatives increase transparency, accountability and access to government services, contribute to better governance and reduce corruption. The use of ICT should be increased in ensuring justice. Accountability can be ensured through e-governance. If proper use of ICT is made in the case of VAT-tax, the government will be able to collect a fair amount of money every year. However, achieving these goals requires addressing significant challenges that undermine the potential of ICT.

Challenges of ICT in Bangladesh

Despite its potential, the effective implementation of ICT in Bangladesh is hampered by the following factors:

Limited connectivity: Although internet penetration has improved, rural areas still suffer from unreliable and slow connections, with only 41% of the population having consistent access.

Power shortage: Frequent power outages disrupt ICT operations, especially in remote areas.

High cost: The high cost of Internet service and digital devices limits access for low-income families.

Digital illiteracy: Many citizens lack basic digital skills, which prevents them from using ICT tools effectively. Educational institutions, especially in rural areas, often lack ICT-trained teachers and adequate resources to teach digital skills.

Policy and regulatory gaps: Implementation Challenges: The ICT Act 2002, aimed at promoting ICT development, has been criticized for lack of transparency in implementation and shortcomings in addressing cyber security concerns.

Cyber Security Risks: Weak regulatory frameworks leave individuals and businesses vulnerable to cyber threats, including data breaches and online fraud.

Urban-rural divide: Urban centers enjoy good ICT infrastructure and services, while rural areas remain underserved.

Gender inequality: Women, especially in rural areas, have limited access to ICT resources due to cultural norms and financial constraints.

Lack of local content and innovation:

Absence of locally relevant digital content in Bengali language limits ICT adoption, especially among non-English speaking populations. We do not invent anything that we can later use to our advantage.

Digital divide in Bangladesh

The digital divide reflects disparities in access and use of ICT among different groups. In Bangladesh, this divide is pronounced across socio-economic, geographic and gender lines. Urban areas such as Dhaka and Chittagong have access to advanced ICT infrastructure, while rural areas face connectivity issues and limited digital services. Only 28% of rural households own a computer, compared to 67% in urban areas. Women are significantly underrepresented in ICT education and employment. According to the GSMA, women in Bangladesh are 33% less likely than men to own a mobile phone or use the internet, largely due to cultural norms and security. High-income groups can afford advanced ICT equipment, while low-income families are excluded from the benefits of digital technology. Schools in rural areas often lack computers, internet connections and trained teachers, leaving students unprepared for the digital economy. Bridging this divide is essential to ensure that ICT contributes to inclusive and equitable development.

Role of ICT Act 2002

In 2002, the first ICT Act was enacted in Bangladesh. Its main objectives include promoting ICT development, facilitating e-governance and ensuring legal framework for electronic transactions. Power of ICT Act 2002. Provides a legal basis for digital transactions, enhancing

e-commerce and online services. And the main goal of this law was to ensure all services by 2006. Even in 2024, the question remains as to how much they have been fulfilled. How applicable this law is in 2024 is also a big question. Facilitate establishment of e-governance platforms, improve public service delivery. Provisions introduced to combat cybercrime and protect digital assets. The law has been criticized for vague definitions and overly broad provisions, particularly regarding the punishment of cybercrimes. Enforcement mechanisms are weak, leading to limited accountability for violations. The absence of strong data protection laws exposes individuals and businesses to cyber security risks.

To address these deficiencies, the government must update the ICT Act to align with emerging digital challenges and international best practices.

ICT opportunities in Bangladesh

Although the development of ICT in Bangladesh is late, there are many job opportunities in the future. There are many sectors of ICT where development is very important. Among these are:

Expansion of digital infrastructure: Nothing is sustainable unless infrastructure is developed. ICT infrastructure is very important in the context of Bangladesh. Because the ICT infrastructure in our country is very fragile. Continued investment in broadband and mobile networks will improve connectivity. Our network is very poor compared to developed countries. Although we say 4G and 5G in our mouth, it is different in reality. As a result, we are far behind the developed countries. For this infrastructure development is very important.

Capacity building: Enhancing digital literacy and skill development programs will empower citizens. Most people in our country do not have digital signatures. They do not know how to use ICT tools. They do not have the ability to understand what is good and what is bad for them. For this, it is necessary to study various types of legal complexities. Steps should be taken by the government to increase digital signature.

New technology: People of our country always have a fear of new technology. People of this country do not want to get used to new technology easily. They think this technology is harmful to them. For example, we were very hesitant about the use of AI. Much later we started using it. Similarly when broadband first came we didn't receive anything. If I had accepted then, maybe I

would not have been so far behind. New technologies can revolutionize sectors such as healthcare, agriculture and finance.

Public-Private Partnerships: Collaboration between the public and private sectors should enhance innovation and efficiency in ICT projects. Through this, as we will get efficient manpower, sustainable development is also possible.

Access to all: ICT devices and internet access should be subsidized for disadvantaged groups. Promotion of this ICT policy is essential for women's empowerment.

Policy reform: The ICT Act 2002 needs to be updated to meet emerging challenges and align with global standards. Alignment between national ICT policies and SDG goals must be ensured.

Strengthening cyber security: Comprehensive cyber security framework and awareness programs should be developed. So that incidents like theft of reserves do not happen again in Bangladesh.

Investment in education and research: The investment in education and research in our country is very low. As a result, researchers do not come out in our country like other countries. ICT should be included in primary and secondary education curriculum. Research centers should be established for innovation in ICT.

Conclusion

ICT is undoubtedly a transformative force for achieving sustainable development in Bangladesh. Its integration in key sectors such as education, healthcare, governance and economy has already shown promising results. Initiatives under the "Digital Bangladesh" vision have expanded digital infrastructure, increased access to technology and fostered innovation. However, significant challenges remain, such as the digital divide, inadequate infrastructure in rural areas, cyber security concerns and skills gaps. These challenges must be systematically addressed to harness the full potential of ICT for development.

The digital divide is particularly pressing as it risks excluding rural populations, women and economically disadvantaged groups from the benefits of ICT. Bridging this divide requires targeted investments in digital literacy, infrastructure expansion and inclusive policies. Additionally, updating the ICT Act 2002 to address emerging issues such as artificial

intelligence, data privacy and cyber security will create a strong legal framework for sustainable ICT growth. ICT plays an important role in achieving the Sustainable Development Goals (SDGs) in Bangladesh by promoting access to quality education, decent work and economic growth and fostering innovation. Its potential to reduce inequality and its importance for strengthening global partnerships for sustainable development should be further highlighted.

Looking ahead, the future of ICT in Bangladesh is promising. Adoption of emerging technologies such as artificial intelligence, blockchain and Internet of Things can revolutionize industries and improve service delivery. Public-private partnerships can drive innovation and efficiency, while green ICT practices can promote environmental sustainability. By aligning ICT strategies with the SDGs and adopting a people-centric approach, Bangladesh can accelerate its journey towards inclusive and equitable development. To conclude, the path to using ICT for sustainable development in Bangladesh is full of challenges, but also full of opportunities. A holistic and inclusive approach—focusing on infrastructure development, policy reform, capacity building and reducing inequality—will ensure that ICT becomes the cornerstone of Bangladesh's sustainable future. With strong political will and collective efforts of all stakeholders, Bangladesh can set a global example in using ICT as a tool for sustainable development, changing the lives of millions and ensuring a brighter future for the nation.

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